C 4573 **(Pages : 3)** **Name.....**

Reg. No.....

SECOND SEMESTER M.Sc. DEGREE (SUPPLEMENTARY) EXAMINATION APRIL 2021

(CUCSS)

Mathematics

MT2C06—ALGEBRA—II

(2016 Admissions)

Time : Three Hours Maximum : 36 Weightage

Part A

Answer **all** *questions.*
Each questions carries 1 *weightage.*

1. Let $R = \left\{ \begin{bmatrix} a & b \\ -b & -a \end{bmatrix} : \det(A) \neq 0 \text{ and } a, b \in \mathbb{R} \right\}$ be a commutative ring with identity prime ideals in R.

2. Is $\sqrt{\pi}$ an algebraic over $\mathbb{Q}(\pi^2)$? Justify your answer.

3. Let $\mathbb{Q}(\sqrt{2}, \sqrt{3}, \sqrt{5})$ be a extension field of $\mathbb{Q}$. Is $\mathbb{Q}(\sqrt{2}, \sqrt{3}, \sqrt{5})$ a simple ? Why ?

4. Find a basis for $\mathbb{Q}(w, i, \sqrt[3]{2})$ over $\mathbb{Q}$, where w is cube root of unity.

5. Find the number of isomorphism of $\mathbb{Q}(\sqrt[4]{2}, i)$ onto to a subfield of $\bar{\mathbb{Q}}$ leaving $\mathbb{Q}$ fixed.

6. Find the number of subfields of a field $\mathbb{Z}_3[x]/\langle x^2 + 1 \rangle$.

7. Find all conjugate of $\sqrt{2 + \sqrt{i}}$ over $\mathbb{Q}$.

8. If α, β are constructible and β is nonzero, then α/β is constructible.

Turn over

9. Is $x^4 + x^2 - 1$ a separable over $\mathbb{Q}$? Why ?

10. Give an example of two finite normal extensions K_1 and K_2 of the same field F such that K_1 and K_2 are not isomorphic fields but $G(K_1/F) \cong G(K_2/F)$.

11. Express $x_1^2 + x_2^2 + x_3^2 + x_4^2$ as polynomials in the elementary symmetric functions in x_1, x_2, x_3, x_4.

12. Prove or disprove : the regular 7-gon is constructible.

13. Prove or disprove : $x^3 - 2$ is solvable by radical over $\mathbb{Q}$.

14. Find $\Phi_{14}(x)$.

(14 × 1 = 14 weightage)

Part B

Answer any **seven** *questions.*
Each questions carries 2 weightage.

15. Find a prime ideal of $\mathbb{Z} \times \mathbb{Z}$ that is not maximal.

16. Let E be an extension of $\mathbb{Z}_2$ and let $\alpha \in E$ be algebraic of degree 3 over $\mathbb{Z}_2$. Classify the groups $(\mathbb{Z}_2(\alpha), +)$ and $\left(\left(\mathbb{Z}_2(\alpha)\right)^*, \cdot\right)$.

17. Prove that an algebraically closed field F has no proper algebraic extension.

18. Determine whether or not the polynomial $x^2 - 6$ is irreducible in $\mathbb{Z}_{11}[x]$.

19. Find all automorphism of $\mathbb{Q}$.

20. Find the splitting field of $x^3 + x^2 + 1$ over $\mathbb{Z}_2$.

21. Describe the group of the polynomial $x^3 - 1 \in \mathbb{Q}[x]$ over Q.

22. Show that in $\mathbb{Q}[x]$, $\Phi_{2n}(x) = \Phi_n(-x)$ for odd integer $n > 1$.

23. Find $\Phi_8(x)$ over $\mathbb{Z}_3$.

24. If α and β are constructible real numbers, prove that $\alpha + \beta$ is constructible.

(7 × 2 = 14 weightage)

Part C

Answer any **two** *questions.*
Each questions carries 4 *weightage.*

25. Prove that every field contains either a subfield is isomorphic to $\mathbb{Z}_p$ for some prime p or a subfield is isomorphic to Q.

26. (a) Show that every finite field is perfect.

    (b) If F is a field, prove that every ideal in F [x] is principal ideal.

27. Find the Galois group of $x^4 - 2$ over $\mathbb{Q}$ and exhibit the correspondence between the subgroups of the Galois group and the intermediate fields.

28. Let F be a field of characteristic zero, and let $a \in F$, If K is a splitting field of $x^n - a$ over F, prove that G (K/F) is a solvable.

(2 × 4 = 8 weightage)